

Align on English anchors only; evaluate on held-out English only

A Pretraining and same-language alignment

Fit Procrustes maps using
3000 general-vocabulary English anchors

Bilingual EN+ZH

EN tokens + ZH tokens
(pretraining mixture)



Monolingual EN

EN tokens
(pretraining mixture)

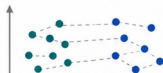


Bilingual EN+FR

EN tokens + FR tokens
(pretraining mixture)



Aligned English subspace
($\text{EN} \leftrightarrow \text{EN+ZH}$)



English-to-English
alignment only



Aligned English subspace
($\text{EN} \leftrightarrow \text{EN+FR}$)



- Bilingual EN+ZH
- Monolingual EN (reference)
- Bilingual EN+FR
- Nearest-neighbor match after alignment

B Held-out English evaluation

Probe: wedding dress
1000 held-out English probes

Axis: red $\leftrightarrow$ white
50 semantic axes

1 Token embeddings

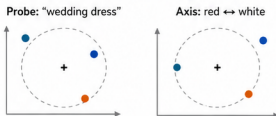
Aligned representations remain close.



Small mismatch after alignment

2 Contextual states

Aligned representations diverge more.



Larger mismatch after alignment

----- Dashed circle denotes average distance from EN reference (smaller is better)